

Def. Let  $F$  be a field. Given a polynomial  $f(x) \in F[x]$  over  $F$ , the Extension  $E \stackrel{\text{def}}{=} F(\alpha_1, \dots, \alpha_n)$  where  $\alpha_1, \dots, \alpha_n \in \bar{F}$  are roots of  $f$ .

is called the splitting field of  $f(x)$ .

Further, the collection of polynomials  $\{f_i(x) \mid i \in I\}$ , the intersection of splitting fields of  $f_i$  is also called the splitting field.

Thm. Let  $\varphi: F \rightarrow F'$  be an isomorphism of fields.

Given  $f(x) = a_n x^n + \dots + a_1 x + a_0 \in F[x]$ , let  $f'(x) = \varphi(a_n) x^n + \dots + \varphi(a_1) x + \varphi(a_0) \in F'[x]$ .

Let  $E$  be the splitting field for  $f(x)$  over  $F$ ,

and  $E'$  be the splitting field for  $f'(x)$  over  $F'$ .

Then,  $\varphi$  can be extended on  $E$  to  $E'$ , isomorphically.

Dummit. sec 13.4.

Prop. Let  $K$  be a finite extension of  $F$ .

$K/F$  is splitting field

if and only if

every irreducible polynomial in  $F[x]$  that has a root in  $K$  splits over  $K$ .

Proof. Suppose that  $K$  is the splitting field of  $\{f_i(x) \mid i \in \{1, 2, \dots, n\}, f_i \in F[x]\}$ . Since  $K/F$  is finite, the collection  $\{f_i\}$  was chosen finite. Simply, we say that  $K$  is a splitting field of  $f = f_1 \cdots f_n \in F[x]$ . 이것이 필요한 가정은 아님.

Let  $p(x) \in F[x]$  be an irreducible polynomial such that  $p(\alpha) = 0$  for some  $\alpha \in K$ . Then, for any other root  $\beta \in F$  of  $p(x)$ , there is a conjugate isomorphism  $\gamma: F(\alpha) \rightarrow F(\beta)$  which fixes  $F$  and maps  $\alpha$  to  $\beta$ .

Now, by the Extension Theorem, there is an extension of  $\gamma$ ,  $\gamma: K \rightarrow F$  such that  $\gamma|_{F(\alpha)} = \gamma$ . But, since  $K/F$  is splitting field and  $\gamma$  fixes  $F$ , so  $\gamma[K] = K$ . Therefore,  $F(\beta) \subseteq K$ , so  $\beta \in K$ .

Conversely, suppose that the below. Since  $K/F$  is finite, we can write  $K = F(\alpha_1, \dots, \alpha_r)$ . Now, let  $f_i \in F[x]$  be a minimal polynomial of  $\alpha_i$ . Let  $L$  be the splitting field of  $f_1, \dots, f_r$  over  $F$ . By the assumption, all  $f_i$  split over  $K$ , so  $L \subseteq K$ . Reversely, each  $\alpha_i$  is a root of  $f_i$ , so  $K = F(\alpha_1, \dots, \alpha_r) \subseteq L$ . Consequently,  $K = L$ .  $\square$

Failed try:  $\Rightarrow$

Let  $f(x) \in F[x]$  be an irreducible polynomial such that for some  $\alpha \in K$ ,  $f(\alpha) = 0$ . If  $\alpha \in F$ , then  $f$  is just degree 1 polynomial, so there is nothing to prove. If  $\alpha \notin F$ , then  $\alpha$  is a root of for some  $f_3$ , and since  $f_3$  splits over  $K$ , so the irreducible  $f$  is also minimal polynomial of  $\alpha$ , so  $f$  divides  $f_3$ , therefore  $f$  splits over  $K$ .

$\alpha$ 가 어떤  $f_3$ 의 근일 필요는 없다!

Ex)  $\mathbb{Q}(\sqrt{2}, \sqrt{3})$  은  $(x^2 - 2)(x^2 - 3)$  의 splitting field.  
하지만  $\sqrt{2} + \sqrt{3}$  의 minimal polynomial은  $x^4 - 10x^2 + 1$ .

Prop. Let  $K_1, K_2$  be finite extensions of  $F$  contained in the field  $K$ , and suppose that  $K_1$  and  $K_2$  both are splitting fields over  $F$ .

Then,  $K_1 K_2$  and  $K_1 \cap K_2$  both are splitting fields over  $F$ .

Proof. Let  $K_1$  and  $K_2$  be splitting field of  $f_1$  and  $f_2$ , respectively.

Then, for all roots  $\alpha_1, \dots, \alpha_k$  and  $\beta_1, \dots, \beta_r$  of  $f_1$  and  $f_2$ , respectively,

we can write  $K_1 = F(\alpha_1, \dots, \alpha_k)$  and  $K_2 = F(\beta_1, \dots, \beta_r)$

Then, the Composition field  $K_1 K_2 = F(\alpha_1, \dots, \alpha_k, \beta_1, \dots, \beta_r)$ .

Meanwhile, let  $L$  be the splitting field of  $f_1 \cdot f_2$ . Then,

$L$  contains all roots of  $f_1 f_2$ , so  $K_1 K_2 \subseteq L$ . And  $L$  is splitting field of  $f_1 f_2$ , so  $L \subseteq K_1 K_2$ , so  $L = K_1 K_2$  is splitting field.

Let  $\alpha \in K_1 \cap K_2 \setminus F$  be given. Then there is minimal irreducible polynomial  $f(x)$  over  $F$  having  $\alpha$  as a root. Since  $\alpha \in K_1$  and  $\alpha \in K_2$ ,

the  $f(x)$  splits in both  $K_1$  and  $K_2$ , so for  $\{\alpha_1, \dots, \alpha_n\}$ , the set of all roots of  $f$ ,  $F(\alpha_1, \dots, \alpha_n) \subseteq K_1 \cap K_2$ , so  $f$  splits over  $K_1 \cap K_2$ .

Since  $\alpha$  was arbitrary, the  $K_1 \cap K_2$  is splitting field.

Def. A field  $K/F$  is called separable if:

every element of  $K$  is the root of a separable polynomial over  $F$ .

Thm 13.5

Prop. Let  $F$  be any field, and  $d, n \in \mathbb{N}$ . Then

$d$  divides  $n$  if and only if  $x^d - 1$  divides  $x^n - 1$ .

Proof. Suppose that  $n = dk$  for some  $k \geq 2$ . (If  $k=1$ , we are done)

$$x^n - 1 = x^{dk} - 1 = (x^d)^k - 1 = (x^d - 1)(x^{d(k-1)} + x^{d(k-2)} + \dots + 1)$$

So,  $x^d - 1$  divides  $x^n - 1$ . Conversely, suppose that  $x^d - 1$  divides  $x^n - 1$ .

By the division algorithm,  $n = qd + r$  for some  $q, r \in \mathbb{Z}$  and  $0 \leq r < d$ .

$$\text{So, } x^n - 1 = x^{qd+r} - x^r + x^r - 1 = x^r(x^{qd} - 1) + x^r - 1. \quad (q \neq 0, \text{ since } n \geq d)$$

Since  $x^{qd} - 1 = (x^d - 1)(x^{d(q-1)} + \dots + x + 1)$  is divided by  $x^d - 1$ ,  
by the assumption,  $x^r - 1$  is divided by  $x^d - 1$ . But, since  $r < d$ ,  
the remainder  $r$  must be zero, so  $n = qd$ .  $\square$

Prop.  $\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^n}$  if and only if  $d$  divides  $n$ .

Proof. Since

$$\mathbb{F}_{p^k} = \{x \in \overline{\mathbb{F}_p} \mid x^{p^k} - x = 0\}$$

and the fact that

$$\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^n} \Leftrightarrow x^{p^d} - x \mid x^{p^n} - x$$

$$\Leftrightarrow p^d - 1 \mid p^n - 1$$

$$\Leftrightarrow d \mid n$$

So proved.

